

Introduction to Machine Learning

with Python & scikit-learn

Course slides based on
Introduction to Machine Learning with Python

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About This Course

- **Goal:** build working machine learning solutions in Python — focus on *practice*, not heavy math
- **Tools:** Python 3, scikit-learn, NumPy, pandas, matplotlib
- **Prerequisites:** basic Python; some NumPy / matplotlib familiarity helps
- **Approach:** every concept paired with runnable code and real datasets

"We focus on the practical aspects of using machine learning algorithms, rather than the math behind them."

About the Teachers



Zhang Li (张丽)

Course Leader

Assistant Professor

Ph.D in Computer Science, Shandong University, China

Overseas Experience in USA

About the Teachers (continued)



Jing Ming (井明)

Theory Lecture

Lecturer

Ph.d in Software Engineering, Shandong University, China

Overseas Experience in Singapore

Course Assessment Methods

Assessment	Weight	Description
Assignments	20%	2 assignments, each worth 10%
Lab Reports	20%	2 lab reports, each worth 10%
Quizzes	10%	2 quizzes, each worth 5%
Project Report	50%	Final project, worth 50%

Course Roadmap

#	Chapter	Theme
1	Introduction	ML concepts, tools, first model
2	Supervised Learning	Classification & regression algorithms
3	Unsupervised Learning	Preprocessing, dimensionality reduction, clustering
4	Data Representation	Feature engineering & selection
5	Model Evaluation	Cross-validation, grid search, metrics
6	Pipelines	Chaining preprocessing + models safely
7	Text Data	Bag-of-words, tf-idf, topic modeling
8	Wrapping Up	Workflow, production, next steps

Let's begin

Chapter 1 → Introduction